Pre-analysis plan for #ManyDaughters

Johanna Sophie Quis*

July 3, 2025

* University of Hannover, Königsworther Platz 1, 30167 Hannover, Germany, phone: +49-511-762-14974, e-mail: johanna.sophie.quis@ihe.uni-hannover.de

1 Introduction

The project #ManyDaughters is a multi-analyst study of the causal effects of having daughters on respondent's attitudes, preferences, and behaviors. This pre-analysis plan thereby outlines the analysis of one research team (RT) among a multitude of research teams, each analyzing four given causal hypotheses based on a 5% sample of the SOEP dataset in the years 2000-2020. The Project Coordinators (PCs) then run the analyses on the remaining 95% of the dataset to gain insights into the research questions as well as the scientific process the RTs chose.

2 Study Desgin

In this section I outline the hypotheses, identification strategy, and the main outcome variables of interest

2.1 Hypotheses

The hypotheses to be tested include

H1: Having daughters leads to stronger support for gender equality.
That is, denoting X as a measure of support for Gender equality,

$$(X\text{for respondents with daughters}) > (X\text{for respondents without daughters})$$

H2: Having daughters leads to more left-leaning political preferences.
That is, denoting Y as a measure of left-leaning political preferences,

$$(Y\text{for respondents with daughters}) > (Y\text{for respondents without daughters})$$

H3: Having daugthers leads to more gender equality in the household (as measured by behaviors or outcomes).
That is, denoting Z as a measure of gender equality in the household,

$$(X\text{for respondents with daughters}) > (X\text{for respondents without daughters})$$

H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).
That is, denoting X as a measure of support for gender equality,

$$(X\text{for male respondents with daughters} - \text{for male respondents without daughters})$$

$$> (X\text{for female respondents with daughters} - X\text{for female respondents without daughters})$$

2.2 Main Identification

I assume that gender composition of an individual's offspring is random to the individual. However, the total number of children might depend partly on the gender composition of the previously born children. Therefore, I control for the total number of children.

I estimate linear regressions (Stata command `regress`) using robust standard errors in all specifications

Model for H1:

$$\text{SupportGE}_i = \beta_0 + \beta_1 \cdot \text{Daughters}_i + X_i' \cdot \gamma + \epsilon_i$$

Where SupportGE_i is the measure of support for gender equality, Daughters_i a binary variable, stating that the household has at least one daughter, and X_i includes the controls (total number of children, income, education, marital status, migration background, male).

To assess H1: I will check if the 95% confidence interval of β_1 is strictly above 0, and reject H1 otherwise

Model for H2:

$$\text{LeftLeaning}_i = \beta_0 + \beta_1 \cdot \text{Daughters}_i + X_i' \cdot \gamma + \epsilon_i$$

Where LeftLeaning_i is the measure of left-leaning political preferences, Daughters_i a binary variable, stating that the respondent has at least one daughter, and X_i includes the controls (total number of children, income, education, marital status, migration background, male).

To assess H2: I will check if the 95% confidence interval of β_1 is strictly above 0, and reject H2 otherwise

Model for H3:

$$\text{ActualGE}_i = \beta_0 + \beta_1 \cdot \text{Daughters}_i + X_i' \cdot \gamma + \epsilon_i$$

Where ActualGE_i is the measure of realized gender equality in the household, Daughters_i a binary variable, stating that the respondent has at least one daughter, and X_i includes the controls (total number of children, income, education, marital status, migration background, male).

To assess H3: I will check if the 95% confidence interval of β_1 is strictly above 0, and reject H3 otherwise

Model for H4:

$$\text{SupportGE}_i = \beta_0 + \beta_1 \cdot \text{Daughters}_i + \beta_2 \cdot \text{Male}_i + \beta_3 \cdot \text{Daughters}_i \times \text{Male}_i + X_i' \cdot \gamma + \epsilon_i$$

Where SupportGE$_i$ is the measure of realized gender equality in the household, Daughters$_i$ a binary variable, stating that the respondent has at least one daughter, and X_i includes the controls (total number of children, income, education, marital status, migration background).

To assess H4: I will check if the 95% confidence interval of β_3 is strictly above 0, and reject H4 otherwise

2.3 Alternative Identification

If possible, I would instrument having daugthers with the gender of the first born child. The identification strategy is also detailed here, however, the applicability depends on the quality of the instrument. If the first stage fails, I will not include the results.

Thus, analog to section 2.2 I will estimate the same equations, but instead of Daughters$_i$ now use $\widehat{\text{Daughters}}_i$ estimated in the following first stage equation:

$$\text{Daughters}_i = \pi_0 + \pi_1 \cdot \text{DaughterFirst}_i + X_i' \cdot \pi_2 + \xi_i$$

Where Daughters$_i$ is the estimated probability of having a daughter in the household, DaughterFirst$_i$ a binary variable, stating that the respondent's firtborn child is a daughter, and X_i includes the controls (total number of children, income, education, marital status, migration background).

The alternative estimations will be done via Stata's `ivregress`-command

2.4 Outcome Variables

This subsection details the outcome variables I selected in order to inspect H1 to H4

SupportGE: Measure of support for gender equality

- Attitudes asked in 8 items (listed below) in question 197 of the 2018 individual questionnaire.
  https://paneldata.org/soep-core/instruments/soep-core-2018-pe-m12/197
- Items are on a 1-7 Lickert Scale, where 1 means completely disagree, 7 means completely agree
  1. *A person who is living with their partner for the long term should get married.* **(excluded)**
  2. I think it is good that marriages between two women or two men are legally recognized.
     Variable: `plh0357`
     https://paneldata.org/soep-core/datasets/pl/plh0357
  3. *A single parent can raise a child just as well as two parents together.* **(excluded)**

4. Children below the age of 6 suffer if their mother works. **(reversed)**
   Variable: `plh0298_v2`
   https://paneldata.org/soep-core/datasets/pl/plh0298_v2
5. A child under the age of three suffers if his or her mother works. **(reversed)**
   Variable: `plh0302_v2`
   https://paneldata.org/soep-core/datasets/pl/plh0302_v2
6. It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally.
   Variable: `plh0308_v2`
   https://paneldata.org/soep-core/datasets/pl/plh0308_v2
7. A same-sex couple can raise a child just as well as a man and woman.
   Variable: `plh0359`
   https://paneldata.org/soep-core/datasets/pl/plh0359
8. It would be beneficial to society to recognize transgender people (people who changed their biological sex) as normal.
   Variable: `plh0360`
   https://paneldata.org/soep-core/datasets/pl/plh0360

- I will generate an index from 6 out of the 8 items, that are contained in question 197/2018, excluded items are presented in italics. Items 4 and 5 will be reversed in order to have higher values represent more support for gender equality.
- The index is constructed as follows: I will add the scores from all 6 items and z-standardize the resulting index to a mean of zero and a standard deviation of 1. I will impute up to two missing values in the index, by using the mean value of the other items from the index. Individuals with more than two missing values will be excluded from the analyses.

LeftLeaning: measure of left-leaning political preferences

- Political attitude on a 0-10 Lickert Scale with 0 (left) and 10 (right), assessed in 2005, 2009, 2014, 2019
  Variable: `plh0004` https://paneldata.org/soep-core/datasets/pl/plh0004
- I will use the 2019 value reversed, such that higher values reflect more left-leaning tendencies

ActualGE: Measure of realized gender equality in the household

- number of hours spent on household and care from question 17 in the 2018 individual questionnaire
- https://paneldata.org/soep-core/instruments/soep-core-2018-pe-m12/17

1. *Work, apprenticeship (including travel time to and from work) [number of hours]*
2. Errands (shopping, trips to government agencies, etc.) [number of hours]
3. Housework (washing / cooking / cleaning) [number of hours]
4. Childcare [number of hours]
5. Care and support of persons in need of care [number of hours]
6. *Education or further training (also school / university) [number of hours]*
7. Repairs on and around the home, car repairs, garden or lawn work [number of hours]
8. *Physical activity (sport, fitness, gymnastics) [number of hours]*
9. *Hobbies and other leisure-time activities [number of hours]*

– I will add the hours from 5 out of the 9 items (excluded items are presented in italics), that are contained in question 17/2018, for both partners in the household. Then, I will calculate the absolute of difference between both partner's hours spent. Thus, the higher the difference, the more unequal is the distribution of care work between the individuals. I use the absolute difference, in order to be able to also include households with same-sex partners or reversed role models (e. g. with a stay at home dad).

2.5 Additional Variables

I need the following additional variables:

- household ID `hhid`
- individual ID `pid`
- survey year `syear`
- partner ID `parid`
- sex of the respondent `sex`, recoded to a binary variable `male` if the respondent is male, 0 otherwise
- number of children the respondent has `sumkids`
- birth year of the children `kidgeb01-kidgeb19` in order to only include children that were already born in 2018/2019
- sex of the children `kidsex01-kidsex19` converted into a binary variable `Daughters` that takes value 1 if person has at least 1 daughter, 0 otherwise

- household income `hghinc`
- education in years `pgbilzeit` - not available for 2018, thus will use the value from 2017 for 2018 analyses
- marital status `partner` included as categorical variable with "spouse" as base category
- migration background `migback`

2.6 Inclusion criteria

The respondents need to live in a household with a partner (in order to analyze H3), as I am using information from 2018 and 2019 in order to define the outcomes, the sample is also restricted to respondents who provided answers on the outcome variables in the respective year. For all other variables I will use the newest available information (but not newer than the outcome measures). I will apply listwise deletion in case of missing values on the variables specified above.